1 Vector Spaces

Definition 1.1 (K-vector space)**.** A *vector space* over the field K consists of a set V, an *addition* $V \times V \to V$ and a *scalar multiplication* $K \times V \to V$ such that $(V, +)$ is an abelian group (associative, commutative, zero vector 0, inverses $-v$) and for all $\alpha, \beta \in K$, $v, w \in V$:

$$1v = v, \quad \alpha(\beta v) = (\alpha\beta)v, \quad \alpha(v+w) = \alpha v + \alpha w,$$
$$(\alpha+\beta)v = \alpha v + \beta v.$$

Proposition 1.1 (Computation rules)**.** *For every $\alpha \in K$ and $v \in V$: $\alpha 0 = 0$ and $0v = 0$; if $\alpha v = 0$ then $\alpha = 0$ or $v = 0$; $(-1)v = -v$.*

Example 1.1. $K^n = K \times \cdots \times K$ with componentwise operations; the polynomial space $K[X]$; the function spaces $C^k(I)$ and the analytic functions $O(I)$ on an interval I; the convergent sequences (x_n) in K; the solutions of a linear differential equation.

Definition 1.2 (Subspace)**.** $T \subseteq V$ is a *(linear) subspace* if addition and scalar multiplication restrict to $T \times T$ and $K \times T$, making T itself a vector space.

Lemma 1.1 (Subspace criterion)**.** *$T \subseteq V$ is a subspace iff $T \neq \emptyset$ and $x, y \in T \Rightarrow x + y \in T$ and $\alpha \in K$, $x \in T \Rightarrow \alpha x \in T$. (Equivalently: $0 \in T$ and T closed under $x + \alpha y$.)*
Sketch. *If $T \neq \emptyset$, pick $x \in T$; then $0 = 0x \in T$. The axioms are inherited from V.*

Proposition 1.2. *The intersection of any family of subspaces of V is a subspace of V. (The union of two subspaces is a subspace only if one contains the other.)*

Definition 1.3 (Linear map)**.** $f : V \to W$ is *linear* if $f(x+y) = f(x)+f(y)$ and $f(\alpha x) = \alpha f(x)$ for all $x, y \in V$, $\alpha \in K$. Then $f(0) = 0$, $f(-x) = -f(x)$, and compositions of linear maps are linear.

Lemma 1.2 (Image and kernel)**.** *For linear $f : V \to W$: if $T \subseteq V$ is a subspace, so is $f(T) \subseteq W$; if $U \subseteq W$ is a subspace, so is $f^{-1}(U) \subseteq V$. In particular*

$$\operatorname{img} f = f(V) \subseteq W,$$
$$\ker f = f^{-1}\{0\} = \{x \in V \mid f(x) = 0\} \subseteq V$$

are subspaces.

Lemma 1.3. *A linear map f is injective iff $\ker f = \{0\}$.*
Sketch. *$f(x) = f(y) \Leftrightarrow x - y \in \ker f$.*

Lemma 1.4. *If the linear map $f : V \to W$ is bijective, then $f^{-1} : W \to V$ is linear.*

Definition 1.4 (Isomorphism)**.** Bijective linear maps are called *isomorphisms*; V and W are *isomorphic* ($V \cong W$) if an isomorphism $V \to W$ exists. Isomorphic spaces are interchangeable for all linear-algebraic purposes.

Example 1.2. Every matrix $a \in \operatorname{Mat}(p \times n, K)$ defines a linear map $K^n \to K^p$, $x \mapsto ax$, with $(ax)_i = \sum_{j=1}^n a_{ij}x_j$. Further examples: the limit map $(x_n) \mapsto \lim x_n$ on convergent sequences; the differential operators $D : C^{k+1}(I) \to C^k(I)$, $f \mapsto f'$; $D : O(I) \to O(I)$; and generally $Df = f^{(n)} + a_{n-1}f^{(n-1)} + \cdots + a_0 f$ with coefficient functions a_j.

2 Bases & Dimension

Definition 2.1 (Linear span)**.** For $v_1, \ldots, v_r \in V$,

$$\operatorname{Lin}(v_1, \ldots, v_r) := \{\textstyle\sum_{i=1}^r \lambda_i v_i \mid \lambda_i \in K\} \subseteq V$$

is the set of *linear combinations*; it is the smallest subspace of V containing $v_1, \ldots, v_r$ (the image of the coefficient map $K^r \to V$, $(\lambda_i) \mapsto \sum \lambda_i v_i$).

Definition 2.2 (Linear independence)**.** The tuple $(v_1, \ldots, v_r)$ is *linearly independent* if $\sum_i \lambda_i v_i = 0$ implies $\lambda_1 = \cdots = \lambda_r = 0$; otherwise *dependent*. (The empty tuple is independent.) Independence means: the coefficient map is injective, i.e. the coefficients of every combination are uniquely determined.

Lemma 2.1. *$(v_1, \ldots, v_r)$ is dependent iff some v_k is a linear combination of the v_i with $i \neq k$.*

Lemma 2.2. *If $(v_1, \ldots, v_r)$ is independent and $w \in V$, then $(v_1, \ldots, v_r, w)$ is dependent iff $w \in \operatorname{Lin}(v_1, \ldots, v_r)$.*

Definition 2.3 (Basis)**.** $(v_1, \ldots, v_n)$ is a *basis* of V if $\operatorname{Lin}(v_1, \ldots, v_n) = V$ and the tuple is independent — equivalently: every $v \in V$ is a *unique* combination $\sum_i \lambda_i v_i$ (the coefficient map is an isomorphism $K^n \cong V$).

Example 2.1. The *standard basis* $(e_1, \ldots, e_n)$ of K^n; in $\mathbb{R}^2$ two non-collinear vectors form a basis (parallelogram construction); $(1, X, X^2, \ldots, X^r)$ is independent in $K[X]$, more generally any family of polynomials of distinct degrees.

Remark 2.1. A tuple containing the zero vector, or two equal vectors, is dependent. Independence is a property of the *tuple*, not of the underlying set.

Theorem 2.1 (Basis extension)**.** *If $(v_1, \ldots, v_r)$ is independent and $\operatorname{Lin}(v_1, \ldots, v_r, w_1, \ldots, w_s) = V$, then $(v_1, \ldots, v_r)$ can be extended to a basis of V by adding suitable of the w_j.*

Theorem 2.2 (Exchange lemma)**.** *If $(v_1, \ldots, v_r)$ and $(w_1, \ldots, w_s)$ are bases of the same space V, then for each i there is j such that $(v_1, \ldots, v_{i-1}, w_j, v_{i+1}, \ldots, v_r)$ is again a basis.*
Sketch. *$v_i \notin \operatorname{Lin}(v_1, \ldots, \widehat{v_i}, \ldots, v_r)$, so some w_j lies outside this span; adjoin it.*

Theorem 2.3. *Any two bases of V have the same number of elements. If this number is $n \in \mathbb{N}$, V is* finite-dimensional *of* dimension $\dim V = n$*; otherwise* $\dim V = \infty$*. Also* $\dim K^n = n$*.*

Corollary 2.1. *In an n-dimensional space: a spanning family has at least n, an independent family at most n members; n given vectors form a basis iff they are independent iff they span. (Neither implication may be reversed for $r \neq n$.)*

Theorem 2.4. *Every subspace U of a finite-dimensional V is finite-dimensional with $\dim U \leq \dim V$, and $\dim U = \dim V$ only for $U = V$; every basis of U extends to a basis of V.*

Definition 2.4 (Product and sum)**.** The *cartesian product* $V \times W$ with componentwise operations is a vector space, $\dim(V \times W) = \dim V + \dim W$. For subspaces $S, T \subseteq V$ the *sum*

$$S + T := \{x + y \mid x \in S,\ y \in T\} \subseteq V$$

is a subspace; e.g. $\operatorname{Lin}(v_1, \ldots, v_r) + \operatorname{Lin}(w_1, \ldots, w_s) = \operatorname{Lin}(v_1, \ldots, v_r, w_1, \ldots, w_s)$.

Theorem 2.5 (Dimension formula for subspaces)**.** *For finite-dimensional subspaces $S, T \subseteq V$:*

$$\dim S + \dim T = \dim(S \cap T) + \dim(S + T).$$

Sketch. *Extend a basis $(u_1, \ldots, u_r)$ of $S \cap T$ to bases (u_i, v_j) of S and (u_i, w_k) of T; then (u_i, v_j, w_k) is a basis of $S + T$.*

Definition 2.5 (Direct sum / complement)**.** The sum $S+T$ is *direct* ($S \oplus T$) if $S \cap T = \{0\}$; if moreover $S \oplus T = V$, then S and T are *complementary* subspaces. Then $S \times T \to V$, $(x,y) \mapsto x+y$, is an isomorphism, and $\dim S + \dim T = \dim V$. Complements exist by the basis extension theorem and are far from unique.

Example 2.2. $C^0(\mathbb{R}) = S \oplus T$ where $S =$ even, $T =$ odd continuous functions: only 0 is both, and

$$f(t) = \tfrac{f(t)+f(-t)}{2} + \tfrac{f(t)-f(-t)}{2} \in S+T.$$

Remark 2.2. $K[X]$ is infinite-dimensional: $(1, X, X^2, \dots)$ is independent. Its subspace of polynomials of degree $\le k$ has dimension $k+1$.

3 Matrices & Coordinates

Definition 3.1 (Basis isomorphism / card)**.** For a basis $v = (v_1, \dots, v_n)$ of V, the coefficient map $v : K^n \to V$, $(\lambda_i) \mapsto \sum \lambda_i v_i$, is an isomorphism (*basis isomorphism*); its inverse $v^{-1} : V \to K^n$ is the *card* (coordinate map) determined by v.

Theorem 3.1. *Two finite-dimensional K-vector spaces are isomorphic iff they have the same dimension.*

Sketch. *$v^{-1}w$ paste cards; conversely an isomorphism carries a basis to a basis.*

Theorem 3.2 (Matrix of a map on K^n)**.** *For every linear map $f : K^n \to K^p$ there is a* unique *matrix $a \in \mathrm{Mat}(p \times n, K)$ with $f(x) = ax$ for all x.*

Proposition 3.1 (Fundamental rule)**.** *The columns of $a \in \mathrm{Mat}(p \times n, K)$ are the images of the standard basis vectors: $ae_j = a_j$ (j-th column). In general, for bases v of V and w of W, the matrix describing $f : V \to W$ is that of the composition*

$$K^n \xrightarrow{v} V \xrightarrow{f} W \xrightarrow{w^{-1}} K^p.$$

Theorem 3.3. *For $\dim V = n$, $\dim W = p$ and fixed bases, $f \mapsto$ (matrix of f) is a linear isomorphism $\mathrm{Hom}(V,W) \cong \mathrm{Mat}(p \times n, K)$; in particular $\dim \mathrm{Hom}(V,W) = \dim V \cdot \dim W$.*

Theorem 3.4 (Extension theorem)**.** *If $(v_1, \dots, v_n)$ is a basis of V and $w_1, \dots, w_n \in W$ are arbitrary vectors, there is exactly one linear map $f : V \to W$ with $f(v_j) = w_j$ for all j.*

Definition 3.2 (Matrix product)**.** For $b \in \mathrm{Mat}(q \times p, K)$, $a \in \mathrm{Mat}(p \times n, K)$:

$$(ba)_{ik} := \sum_{j=1}^{p} b_{ij} a_{jk} \in \mathrm{Mat}(q \times n, K).$$

Associative and distributive wherever defined; $\lambda(ab) = (\lambda a)b = a(\lambda b)$.

Lemma 3.1. *If f and g are described by matrices a and b (suitable bases), then $g \circ f$ is described by the product ba* — reversed order.

Definition 3.3 (Invertible matrices)**.** $u \in \mathrm{Mat}(p \times n, K)$ is *invertible* if v with $uv = \mathbf{1}_p$ and $vu = \mathbf{1}_n$ exists; then $v = u^{-1}$ is unique. Invertibility forces $p = n$. u is invertible iff $u : K^n \to K^p$ is an isomorphism, and u^{-1} is the inverse map.

Definition 3.4 (General linear group)**.**

$$\mathrm{GL}(n,K) := \{u \in \mathrm{Mat}(n \times n, K) \mid u \text{ invertible}\}$$

is a group under multiplication (non-abelian for $n > 1$); likewise $\mathrm{GL}(V) = \mathrm{Aut}(V)$ for a vector space V.

Definition 3.5 (Elementary matrices)**.** For $n \in \mathbb{N}$ and $k \neq l$: p_{kl} (swap rows k,l; permutation matrix), $d_k(\lambda)$ (multiply row k by λ), $u_{kl}(\lambda)$ (add λ times row l to row k) all lie in $\mathrm{GL}(n,K)$. Multiplying a from the *left* performs the corresponding *elementary row operation*; from the right, the analogous *column operation* (with the roles of k,l interchanged for u_{kl}).

4 Rank & Gaussian Elimination

Definition 4.1 (Rank)**.** $f : V \to W$ has *finite rank* if $\mathrm{img}\, f$ is finite-dimensional; then $\mathrm{rk}\, f := \dim \mathrm{img}\, f$.

Theorem 4.1 (Rank–nullity formula)**.** *For V finite-dimensional and $f : V \to W$ linear:*

$$\dim \ker f + \mathrm{rk}\, f = \dim V.$$

Sketch. *Extend a basis $(v_1, \dots, v_k)$ of $\ker f$ to a basis of V; the restriction of f to $\mathrm{Lin}(v_{k+1}, \dots, v_n)$ is an isomorphism onto $\mathrm{img}\, f$.*

Lemma 4.1 (Invariances)**.** *If g, h are isomorphisms, $\mathrm{rk}(hfg) = \mathrm{rk}\, f$; rank does not depend on the bases chosen. Elementary row and column operations preserve the rank;* row *operations preserve* $\ker a$, column *operations preserve* $\mathrm{img}\, a$.

Proposition 4.1. $\mathrm{img}\, a = \mathrm{Lin}(a_1, \dots, a_n)$ *(column span), so $\mathrm{rk}\, a$ is the dimension of the column span, and $\mathrm{rk}\, a = \mathrm{rk}\, a^t$.*

Definition 4.2 (Echelon form)**.** A matrix is in *row echelon form* if in each row the first nonzero entry (the "step") stands strictly to the right of the step of the row above; zero rows come last. In the *refined* form each step column contains a standard basis vector e_s.

Theorem 4.2 (Gaussian algorithm)**.** *Row operations bring any matrix into row echelon form; further row operations refine it. (Column echelon forms are defined analogously and reached by column operations.)*

Lemma 4.2 (Reading off image and kernel)**.**

- *If a has* column *echelon form, its nonzero columns form a basis of $\mathrm{img}\, a$; the number of steps is $\mathrm{rk}\, a$.*
- *If a has* refined row *echelon form with step indices $j_1 < \cdots < j_r$, then a basis of $\ker a$ is given by the vectors v_k (k not a step index) with $(v_k)_k = 1$, $(v_k)_{j_s} = -a_{sk}$, all other entries 0.*

Proposition 4.2 (Computing)**.**

- *Basis of $\mathrm{Lin}(v_1, \dots, v_n)$ for $v_j \in K^p$: put the v_j as columns into a matrix, apply the Gaussian algorithm; either read off the nonzero columns (column version) or select those original columns that stand at the step indices (row version). Then the span has dimension $r = \mathrm{rk}\, a$, and $(v_1, \dots, v_n)$ is independent iff $r = n$.*
- *Basis of $S+T$: apply this to the union of the two spanning families.*

Theorem 4.3 (Rank normal form)**.** *For $f : V \to W$ ($\dim V = n$, $\dim W = p$) there are bases of V and W with respect to which f is described by*

$$\begin{pmatrix} \mathbf{1}_r & 0 \\ 0 & 0 \end{pmatrix} \in \mathrm{Mat}(p \times n, K), \qquad r = \mathrm{rk}\, f.$$

5 Linear Equations & Matrix Inversion

Definition 5.1 (Affine subspace). $A = a + L := \{a + x \mid x \in L\}$ with $a \in V$ and a subspace $L \subseteq V$ is an *affine subspace*; $\emptyset$ is affine by convention.

Lemma 5.1. *For a nonempty affine subspace $A = a + L$: the parallel subspace $L = \{x - y \mid x, y \in A\}$ is determined by A; define $\dim A := \dim L$. Moreover $A = b + L$ iff $b \in A$; the affine subspaces containing 0 are exactly the linear subspaces.*

Lemma 5.2 (Fibres). *For linear $f : V \to W$ and $b \in W$, the fibre $f^{-1}\{b\}$ is empty or an affine subspace with parallel space $\ker f$: if $f(a) = b$ then $f^{-1}\{b\} = a + \ker f$.*

Proposition 5.1 (Solving $ax = b$). *Apply row operations to $(a \mid b)$ until the left block has echelon form; read off the solution set $x_0 + \ker a$. Consistency: solvable iff the ranks of a and $(a|b)$ agree.*

Theorem 5.1 (Invertibility criteria). *For $a \in \mathrm{Mat}(n \times n, K)$ the following are equivalent:*

- $\mathrm{rk}\, a = n$ *(a surjective);*
- $\ker a = \{0\}$ *(a injective);*
- $a \in \mathrm{GL}(n, K)$*;*
- *there is x with $ax = \mathbf{1}$;*
- *there is x with $xa = \mathbf{1}$.*

Then necessarily $x = a^{-1}$.

Sketch. *Rank–nullility: injective ⇔ surjective for square matrices; then the left/right inverse is the inverse.*

Proposition 5.2 (Inverting by elimination). *Row-reduce $(a \mid \mathbf{1}) \in \mathrm{Mat}(n \times 2n, K)$: if the left block reaches echelon form with n steps, the refinement yields $(\mathbf{1} \mid a^{-1})$; fewer steps mean a is not invertible.*

Example 5.1. $a = \left(\begin{smallmatrix} 1 & 1 \\ 2 & 3 \end{smallmatrix}\right)$: row-reducing $(a \mid \mathbf{1})$ gives $a^{-1} = \left(\begin{smallmatrix} 3 & -1 \\ -2 & 1 \end{smallmatrix}\right)$.

6 Linear Differential Equations

Definition 6.1. With continuous coefficients $a_0, \dots, a_{n-1}$ and b on an interval $I \subset \mathbb{R}$,

$$Df = f^{(n)} + a_{n-1} f^{(n-1)} + \cdots + a_1 f' + a_0 f = b$$

is a *linear differential equation*; $D : C^n(I) \to C^0(I)$ is linear, and solving means computing $\ker D$ resp. the fibres $D^{-1}\{b\} = f_0 + \ker D$.

Theorem 6.1 (Existence and uniqueness). *For each point $t_0 \in I$ and vector $(x_0, \dots, x_{n-1}) \in \mathbb{C}^n$ there is exactly one $f \in C^n(I)$ with $Df = b$ and $f^{(j)}(t_0) = x_j$ for $j = 0, \dots, n-1$. Hence the solution space of the homogeneous equation $Df = 0$ has dimension n (the initial value map is an isomorphism).*

Theorem 6.2 (Constant coefficients). *Let $a_j \in \mathbb{C}$ be constants and let $p(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0 = \prod_{j=1}^{r}(X - c_j)^{e_j}$ be the factorization of the* characteristic polynomial. *A basis of the solution space of $Df = 0$ is given by the n functions*

$$t \mapsto t^{\ell} e^{c_j t}, \qquad j = 1, \dots, r, \quad \ell = 0, \dots, e_j - 1.$$

Example 6.1 (Damped harmonic oscillator). $f'' + 2\alpha f' + \beta^2 f = 0$ (α damping, β restoring force): $p(X) = (X + \alpha)^2 + (\beta^2 - \alpha^2)$. For weak damping $\alpha < \beta$ the complex roots $-\alpha \pm i\sqrt{\beta^2 - \alpha^2}$ yield the real solutions $e^{-\alpha t}\cos(\sqrt{\beta^2 - \alpha^2}\, t)$, $e^{-\alpha t}\sin(\sqrt{\beta^2 - \alpha^2}\, t)$ (see Section 9).

7 Determinants

Theorem 7.1 (Definition of the determinant). *There is exactly one map $\det : \mathrm{Mat}(n \times n, K) \to K$ with:*

- $\det$ *is linear in each column separately (multilinearity);*
- $\det a = 0$ *if a has two equal columns (alternating);*
- $\det \mathbf{1} = 1$ *(normalized).*

Lemma 7.1 (Behavior under column operations).

$$\det(ap_{kl}) = -\det a, \qquad \det(ad_k(\lambda)) = \lambda \det a,$$
$$\det(au_{kl}(\lambda)) = \det a,$$

and correspondingly for row operations. Adding to one column a multiple of another never changes the determinant; a zero column forces $\det a = 0$.

Theorem 7.2 (Invertibility test). $\det a \neq 0 \iff a \in \mathrm{GL}(n, K)$.

Sketch. *Bring a to echelon form: the result is triangular with diagonal product $= \pm \det a$; it is $\mathbf{1}$ iff a is invertible.*

Proposition 7.1 (Laplace expansion). *Let a^{kl} be the minor obtained by deleting row k and column l. Then*

$$\det a = \sum_{l=1}^{n} (-1)^{k+l} a_{kl} \det a^{kl} \quad \textit{(along row } k\textit{)},$$

$$\det a = \sum_{k=1}^{n} (-1)^{k+l} a_{kl} \det a^{kl} \quad \textit{(along column } l\textit{)}.$$

Definition 7.1 (Transpose). $(a^t)_{ij} = a_{ji}$. Rules: $(a + b)^t = a^t + b^t$, $(ab)^t = b^t a^t$, $\mathrm{rk}\, a = \mathrm{rk}\, a^t$, and $(a^t)^{-1} = (a^{-1})^t$.

Theorem 7.3. $\det a^t = \det a$*; everything said about columns holds for rows too.*

Theorem 7.4 (Product formula). $\det(ab) = \det a \cdot \det b$. *Thus $\det : \mathrm{GL}(n, K) \to K \setminus \{0\}$ is a group homomorphism; its kernel*

$$\mathrm{SL}(n, K) := \{u \in \mathrm{GL}(n, K) \mid \det u = 1\}$$

is the special linear group. *Also $\det a^{-1} = (\det a)^{-1}$.*

Definition 7.2 (Adjugate). For $a \in \mathrm{Mat}(n \times n, K)$ let $\tilde{a}_{kl} := (-1)^{k+l} \det a^{lk}$ (note the swapped indices) define the *adjugate* $\tilde{a}$.

Theorem 7.5. $a\tilde{a} = \tilde{a}a = \det a \cdot \mathbf{1}$*; hence for invertible a:*

$$a^{-1} = \tfrac{1}{\det a}\,\tilde{a}, \qquad x = \tfrac{1}{\det a}\,\tilde{a}b \quad \textit{(Cramer's rule).}$$

Example 7.1. $\det\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = ad - bc$ and $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)^{-1} = \frac{1}{\alpha\delta - \beta\gamma}\left(\begin{smallmatrix} \delta & -\beta \\ -\gamma & \alpha \end{smallmatrix}\right)$.

Proposition 7.2. *For a block triangular matrix $A = \left(\begin{smallmatrix} a & 0 \\ c & d \end{smallmatrix}\right)$ with square blocks a, d: $\det A = \det a \cdot \det d$.*

Example 7.2 (Vandermonde). $\det\left(\lambda_i^{j-1}\right)_{i,j=1..n} = \prod_{1 \le i < j \le n}(\lambda_j - \lambda_i)$; in particular it vanishes iff two λ_i coincide.

8 Permutations & the Leibniz Formula

Definition 8.1 (Symmetric group). Sym_n = bijections $\sigma : \{1, \dots, n\} \to \{1, \dots, n\}$, with composition, is the *symmetric group*. Cycles $(j_1\, j_2 \dots j_r)$ rotate their entries and fix the rest; cycles moving disjoint entries commute, and every permutation is a product of such cycles. Transpositions $(k\, l)$ swap two entries. $|\mathrm{Sym}_n| = n!$.

Example 8.1. $\mathrm{Sym}_3 = \{1, (1\,2), (1\,3), (2\,3), (1\,2\,3), (1\,3\,2)\}$; the permutation matrix of $(k\, l)$ is the elementary matrix p_{kl}.

Definition 8.2 (Signum)**.** Every σ is a product of transpositions, and the parity of the number of factors is well defined; $\operatorname{sgn}\sigma := (-1)^{\sigma} \in \{\pm 1\}$ is $+1$ for an *even*, -1 for an *odd* permutation. The map $\operatorname{sgn} : \mathrm{Sym}_n \to \{\pm 1\}$ is a group homomorphism.

Theorem 8.1 (Leibniz formula)**.**

$$\det a = \sum_{\sigma \in \mathrm{Sym}_n} \operatorname{sgn}(\sigma)\, a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}.$$

Of theoretical value ($n!$ terms); for computations use elimination or Laplace expansion.

9 Real & Complex Vector Spaces

Proposition 9.1 (Realification)**.** *A complex vector space V becomes a real one $V_{\mathbb{R}}$ by restricting scalars; multiplication by* i *is then a* linear map $h : V_{\mathbb{R}} \to V_{\mathbb{R}}$ *with* $h^2 = -\mathrm{id}$. *For $v_1, \dots, v_r \in V$:*

$$\mathrm{Lin}_{\mathbb{C}}(v_1, \dots, v_r) = \mathrm{Lin}_{\mathbb{R}}(v_1, \dots, v_r, \mathrm{i}v_1, \dots, \mathrm{i}v_r),$$

$(v_1, \dots, v_r)$ is $\mathbb{C}$-independent iff $(v_1, \dots, v_r, \mathrm{i}v_1, \dots, \mathrm{i}v_r)$ is $\mathbb{R}$-independent; hence $\dim_{\mathbb{R}} V = 2 \dim_{\mathbb{C}} V$ for finite-dimensional V.

Lemma 9.1. *If $c = a + \mathrm{i}b \in \mathrm{Mat}(n \times n, \mathbb{C})$ (a, b real) describes $f : \mathbb{C}^n \to \mathbb{C}^n$, then the realified map $f_{\mathbb{R}}$ is described by the real block matrix*

$$c_{\mathbb{R}} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \in \mathrm{Mat}(2n \times 2n, \mathbb{R})$$

with respect to the basis $(v_1, \dots, v_n, \mathrm{i}v_1, \dots, \mathrm{i}v_n)$.

Definition 9.1 (Complexification)**.** For a real vector space V, the *complexification* $V_{\mathbb{C}}$ is $V \times V$ with componentwise addition and

$$(\alpha + \mathrm{i}\mu)(u, v) := (\alpha u - \mu v,\ \mu u + \alpha v).$$

Identifying $u \equiv (u, 0)$ gives $V_{\mathbb{C}} = V \oplus \mathrm{i}V$.

Lemma 9.2. *Every basis of V is a basis of $V_{\mathbb{C}}$ (over $\mathbb{C}$); $(v_1, \dots, v_r)$ is independent in V iff independent in $V_{\mathbb{C}}$; $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V$. Example: $C^k(I, \mathbb{C})$ is the complexification of $C^k(I, \mathbb{R})$.*

Proposition 9.2 (Real solutions from complex ones)**.** *Let D be a linear differential operator (or any real map) and $f = u + \mathrm{i}v$ a complex solution with real u, v. Then u and v are real solutions; if $c = \alpha + \mathrm{i}\beta$ is a non-real root of a real characteristic polynomial, so is $\bar{c}$, and*

$$e^{ct} = e^{\alpha t}(\cos \beta t + \mathrm{i} \sin \beta t)$$

yields the real basis solutions $e^{\alpha t} \cos \beta t$ and $e^{\alpha t} \sin \beta t$.

10 Eigenvalues & Diagonalization

Definition 10.1 (Endomorphism)**.** Linear maps $f : V \to V$ are called *endomorphisms* ($\mathrm{End}(V)$); bijective ones *automorphisms* ($\mathrm{Aut}(V)$).

Definition 10.2 (Eigenvalues)**.** $\lambda \in K$ is an *eigenvalue* of $f \in \mathrm{End}(V)$ if some $v \neq 0$ has $f(v) = \lambda v$; such v is an *eigenvector*. The *eigenspace*

$$E_\lambda = \{v \in V \mid f(v) = \lambda v\} = \ker(f - \lambda \,\mathrm{id})$$

is a subspace containing 0 (which does not count as an eigenvector).

Definition 10.3 (Characteristic polynomial)**.** For $a \in \mathrm{Mat}(n \times n, K)$:

$$\chi_a(X) := \det(X\mathbf{1} - a) \in K[X]$$

is monic of degree n. Writing $\chi_a = X^n + c_{n-1}X^{n-1} + \cdots + c_0$:

$$c_0 = (-1)^n \det a, \qquad -c_{n-1} = \sum_{j=1}^{n} a_{jj} := \operatorname{tr} a.$$

Lemma 10.1. *χ_a does not depend on the basis: if a, b describe the same $f \in \mathrm{End}(V)$ in different bases, $\chi_a = \chi_b$. Thus $\det f$, $\operatorname{tr} f$, χ_f are well defined for endomorphisms of finite-dimensional spaces.*

Theorem 10.1. *$\lambda \in K$ is an eigenvalue of f iff $\chi_f(\lambda) = 0$: the eigenvalues are the roots of the characteristic polynomial (in K; complex roots may not exist over $K = \mathbb{R}$).*

Example 10.1. $a = \left(\begin{smallmatrix} 1 & 2 \\ 4 & 3 \end{smallmatrix}\right)$: $\chi_a = (X-1)(X-3) - 8 = (X+1)(X-5)$, eigenvalues -1 and 5. The rotation $a = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ has $\chi_a = X^2 + 1$: no real eigenvalues; after complexification $\mathbb{C}^2$ splits into eigenspaces for i and $-\mathrm{i}$.

Theorem 10.2 (Eigenspaces are independent)**.** *Eigenvectors belonging to pairwise distinct eigenvalues are linearly independent; equivalently the sum $E_{\lambda_1} + \cdots + E_{\lambda_r}$ of eigenspaces for distinct eigenvalues is direct.*

Sketch. *Induction on r: apply f to and subtract λ_r times a dependence relation, then use independence of the first $r-1$ eigenvectors.*

Definition 10.4 (Diagonalizable)**.** f is *diagonalizable* if V is the (direct) sum of the eigenspaces of f. Choosing a basis of eigenvectors (*eigenbasis*), f is described by the diagonal matrix $\mathrm{diag}(\lambda_1, \dots, \lambda_n)$ — each basis vector is merely scaled.

Theorem 10.3 (Trace and determinant via eigenvalues)**.** *If χ_f splits into linear factors, $\chi_f(X) = \prod_{j=1}^{n}(X - \lambda_j)$, then*

$$\operatorname{tr} f = \sum_{j=1}^{n} \lambda_j, \qquad \det f = \prod_{j=1}^{n} \lambda_j.$$

Theorem 10.4 (Triangular form)**.** *If χ_f splits into linear factors, there is a basis with respect to which f is described by an upper triangular matrix; the diagonal entries are the eigenvalues (with multiplicity). Conceptually: there is a* flag $\{0\} = V_0 \subset V_1 \subset \cdots \subset V_n = V$, $\dim V_k = k$, *invariant under f.*

Sketch. *Induction on n: pick an eigenvector v_1, extend to a basis, and apply the induction hypothesis to the induced endomorphism of $V/\operatorname{Lin}(v_1)$.*

Theorem 10.5 (Multiplicities)**.** *If λ is a root of χ_f of multiplicity $e > 0$ (algebraic multiplicity), then $1 \le \dim E_\lambda \le e$ (geometric multiplicity).*

Corollary 10.1. *If χ_f splits into n* distinct *linear factors, f is diagonalizable. If f is diagonalizable, its invariant subspaces are exactly the sums $U_1 \oplus \ldots \oplus U_r$ with $U_i \subseteq E_{\lambda_i}$.*

Example 10.2. A diagonalizable f with only non-negative eigenvalues has a commuting square root: on an eigenbasis set $h(v_j) = \sqrt{\lambda_j}\, v_j$; then $h \circ h = f$ and $h \circ f = f \circ h$.

Remark 10.1 (Nilpotent endomorphisms). For $f \in \mathrm{End}(V)$, $\dim V < \infty$, the following are equivalent: f is nilpotent (some $f^r = 0$); 0 is the only eigenvalue; some basis describes f by a strictly upper triangular matrix.

11 Euclidean Vector Spaces

Definition 11.1 (Symmetric bilinear form)**.** A map $\langle\cdot,\cdot\rangle : V \times V \to K$ that is linear in each variable and symmetric is a *symmetric bilinear form*. Every symmetric matrix $s \in \mathrm{Sym}(n, K)$ ($s^t = s$) defines one on K^n via $\langle x, y\rangle = x^t s y$, and on a general V every basis identifies forms with symmetric matrices.

Definition 11.2 (Inner product, Euclidean space)**.** A symmetric bilinear form on a *real* space V is an *inner product* (Skalarprodukt) if it is positive definite: $\langle v, v\rangle > 0$ for $v \neq 0$. A real space with inner product is a *Euclidean vector space*; the *norm* is $\|v\| \coloneqq \sqrt{\langle v, v\rangle}$.

Theorem 11.1 (Cauchy–Schwarz inequality)**.**

$$|\langle v, w\rangle| \leq \|v\| \cdot \|w\|,$$

with equality iff v, w are linearly dependent.

Sketch. *For $w \neq 0$ set $\lambda \coloneqq \langle v, w\rangle / \langle w, w\rangle$ and expand $0 \leq \langle v - \lambda w, v - \lambda w\rangle$.*

Corollary 11.1 (Norm properties)**.** $\|v\| \geq 0$, $\|v\| = 0 \Leftrightarrow v = 0$; $\|\alpha v\| = |\alpha|\|v\|$; *triangle inequality* $\|v \pm w\| \leq \|v\| + \|w\|$.

Example 11.1. The standard inner product on $\mathbb{R}^n$: $\langle x, y\rangle = x^t y = \sum_j x_j y_j$ (matrix $\mathbf{1}$). Unless stated otherwise, $\mathbb{R}^n$ carries this product.

Definition 11.3 (Angle, orthogonality)**.** For $v, w \neq 0$ the angle between them is $t = \arccos \frac{\langle v, w\rangle}{\|v\|\|w\|} \in [0, \pi]$. Vectors are *orthogonal* ($v \perp w$) if $\langle v, w\rangle = 0$ (the zero vector is orthogonal to everything). A tuple $(v_1, \ldots, v_r)$ is an *orthonormal system* if $\langle v_j, v_k\rangle = \delta_{jk}$.

Lemma 11.1. *Every orthonormal system is linearly independent. If $(v_1, \ldots, v_n)$ is an* orthonormal basis *(ONB), every $v \in V$ has the Fourier expansion*

$$v = \sum_{j=1}^{n} \langle v_j, v\rangle\, v_j.$$

Theorem 11.2 (Gram–Schmidt orthonormalization)**.** *Let $\{0\} = V_0 \subset V_1 \subset \cdots \subset V_n = V$ be a flag and $(v_1, \ldots, v_r)$ an orthonormal tuple with $\mathrm{Lin}(v_1, \ldots, v_k) = V_k$ for $k \leq r$. It extends to an ONB with $\mathrm{Lin}(v_1, \ldots, v_k) = V_k$ for all k. Practically, given any basis $(w_1, \ldots, w_n)$, replace iteratively*

$$v_k \coloneqq \frac{w_k - \sum_{j<k} \langle v_j, w_k\rangle v_j}{\|w_k - \sum_{j<k} \langle v_j, w_k\rangle v_j\|}.$$

Corollary 11.2. *Every finite-dimensional Euclidean space has an ONB.*

Definition 11.4 (Isometry)**.** A linear $f : V \to W$ between Euclidean spaces is an *isometry* if $\langle f(v), f(w)\rangle = \langle v, w\rangle$ for all v, w. Isometries are injective; f is an isometry iff it sends some (every) ONB of V to an orthonormal system of W. Hence every n-dimensional Euclidean space is isometric to the standard $\mathbb{R}^n$: $\mathbb{R}^n$ is the standard model.

12 Orthogonal Maps & Complements

Definition 12.1. An isometric automorphism of a Euclidean space V is an *orthogonal map*; these form the *orthogonal group* $\mathrm{O}(V) \subseteq \mathrm{GL}(V)$, written $\mathrm{O}(n)$ for $\mathbb{R}^n$. Orthogonal maps preserve lengths and angles.

Proposition 12.1 (Characterizations)**.** *For $u \in \mathrm{Mat}(n \times n, \mathbb{R})$ equivalent: u is orthogonal; the columns of u form an orthonormal system; $u^t u = \mathbf{1}$; u is invertible with $u^{-1} = u^t$; $u u^t = \mathbf{1}$; the rows form an orthonormal system.*

Definition 12.2. The *special orthogonal group* is $\mathrm{SO}(n) = \{u \in \mathrm{O}(n) \mid \det u = 1\} = \mathrm{O}(n) \cap \mathrm{SL}(n, \mathbb{R})$; the orthogonal matrices with $\det = -1$ do not form a group.

Example 12.1. $\mathrm{SO}(2)$ consists exactly of the rotation matrices

$$\begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}, \qquad t \in \mathbb{R}.$$

Definition 12.3 (Orthogonal complement)**.** For a subspace U of a Euclidean space V:

$$U^\perp \coloneqq \{v \in V \mid \langle v, u\rangle = 0 \text{ for all } u \in U\}.$$

Theorem 12.1. *For finite-dimensional V and subspaces S, T, U:*

- *$U^\perp$ is a complement of U: $U \cap U^\perp = \{0\}$, $U + U^\perp = V$, hence $\dim U + \dim U^\perp = \dim V$;*
- *$U^{\perp\perp} = U$;*
- *$(S + T)^\perp = S^\perp \cap T^\perp$ and $(S \cap T)^\perp = S^\perp + T^\perp$.*

Lemma 12.1. *If the subspaces $U_1, \ldots, U_r$ are pairwise orthogonal ($\langle u_i, u_j\rangle = 0$ for $u_i \in U_i$, $u_j \in U_j$, $i \neq j$), their sum is direct.*

Sketch. *Take the inner product of a dependence $\sum u_j = 0$ with u_i.*

Theorem 12.2 (Rotations of space have an axis)**.** *Every $u \in \mathrm{SO}(3)$ has a fixed vector $a \in \mathbb{R}^3 \setminus \{0\}$ (eigenvalue 1): u is a rotation about the axis $\mathbb{R}a$.*

Sketch. *$\det u = 1$ is the product of the (real or conjugate-complex) eigenvalues; an odd-degree real polynomial has a real root, and a conjugate pair would contribute product $|\lambda|^2 = 1$ leaving the third root equal to 1.*

Remark 12.1 (Orthogonal projection). For a subspace U with ONB $(v_1, \ldots, v_r)$, the map $p_U(v) \coloneqq \sum_j \langle v_j, v\rangle v_j$ is the *orthogonal projection* onto U: $p_U^2 = p_U$, $\mathrm{img}\, p_U = U$, $\ker p_U = U^\perp$, and $p_U(v)$ is the point of U closest to v.

13 Dual Spaces & Adjoints

Definition 13.1 (Dual space)**.** $V^* \coloneqq \mathrm{Hom}(V, K) = \{\text{linear } \varphi : V \to K\}$ with pointwise operations is the *dual space*; its elements are *linear forms* (functionals). With respect to a basis $u = (u_1, \ldots, u_n)$, linear forms correspond to rows $z \in \mathrm{Mat}(1 \times n, K)$ via $\varphi(x) = z\, u^{-1}(x)$.

Definition 13.2 (Dual basis)**.** The *dual basis* $u^* = (u^1, \ldots, u^n)$ of V^* is defined by $u^i(u_j) = \delta_{ij}$. If z is the row of φ in the basis u, then z^t is the column of φ in the dual basis u^*.

Definition 13.3 (Dual map)**.** For linear $f : V \to W$ the *dual map* $f^* : W^* \to V^*$ is $f^*(\varphi) = \varphi \circ f$. Rules: $\mathrm{id}^* = \mathrm{id}$, $(g \circ f)^* = f^* \circ g^*$ (order reversal); the matrix of f^* in the dual bases is the transpose of the matrix of f.

Lemma 13.1 (Bidual)**.** *The canonical map $V \to V^{**}$, $v \mapsto (\varphi \mapsto \varphi(v))$, is injective, and for finite-dimensional V an isomorphism.*

Definition 13.4 (Annihilator)**.** For $A \subseteq V$, $\Psi \subseteq V^*$:

$$\mathrm{Ann}\, A \coloneqq \{\varphi \in V^* \mid \varphi(v) = 0\ \forall v \in A\},$$

$$\mathrm{Ann}\, \Psi \coloneqq \{v \in V \mid \varphi(v) = 0\ \forall \varphi \in \Psi\} = \bigcap_{\varphi \in \Psi} \ker \varphi.$$

Lemma 13.2. *For finite-dimensional V:* $\operatorname{Ann} A$ *is a subspace depending only on* $\operatorname{Lin}(A)$*; for subspaces U:*

$$\dim U + \dim \operatorname{Ann} U = \dim V, \qquad \operatorname{Ann}\operatorname{Ann} U = U.$$

Proposition 13.1. $\operatorname{rk} f^* = \operatorname{rk} f$ *and (finite-dimensional case)*

$$\operatorname{img} f^* = \operatorname{Ann}\ker f, \qquad \ker f^* = \operatorname{Ann}\operatorname{img} f.$$

Example 13.1 (Describing a span by equations)**.** Given $U = \operatorname{Lin}(v_1, \dots, v_r) \subseteq K^n$, put the v_j as columns into $a \in \operatorname{Mat}(n \times r, K)$ and solve $za = 0$ for rows z; a basis $z_1, \dots, z_m$ of the solution space yields

$$U = \{x \in K^n \mid z_1 x = \cdots = z_m x = 0\}.$$

Theorem 13.1 (Riesz isomorphism)**.** *For a Euclidean space V, the map $V \to V^*$, $v \mapsto \langle v, \cdot\rangle$, is an isomorphism; its matrix with respect to a basis u and the dual basis u^* is the Gram matrix s of the inner product.*

Definition 13.5 (Adjoint map)**.** For linear $f : V \to W$ between Euclidean spaces, the *adjoint* $f^\dagger : W \to V$ is defined by

$$\langle f(v), w\rangle = \langle v, f^\dagger(w)\rangle \quad \text{for all } v \in V,\ w \in W.$$

Proposition 13.2 (Rules for the adjoint)**.** $\mathrm{id}^\dagger = \mathrm{id}$*;* $(g \circ f)^\dagger = f^\dagger \circ g^\dagger$*;* $(f^\dagger)^\dagger = f$*;* $\operatorname{rk} f^\dagger = \operatorname{rk} f$*;*

$$\operatorname{img} f^\dagger = (\ker f)^\perp, \qquad \ker f^\dagger = (\operatorname{img} f)^\perp,$$

so that $V = \ker f \oplus \operatorname{img} f^\dagger$ and $W = \operatorname{img} f \oplus \ker f^\dagger$ are orthogonal decompositions.

Definition 13.6 (Hermitian forms, unitary spaces)**.** On a complex space, a *Hermitian form* $\langle\cdot,\cdot\rangle : V \times V \to \mathbb{C}$ is linear in the second and conjugate-linear in the first variable, with $\langle v, w\rangle = \overline{\langle w, v\rangle}$. The standard example on $\mathbb{C}^n$ is $\langle w, z\rangle = \overline{w}^t z$. A positive definite Hermitian form makes V a *unitary space*; matrices representing it satisfy $\bar{s}^t = s$.

14 Normal Endomorphisms & Spectral Theorems

Definition 14.1 (Normal)**.** An endomorphism f of a Euclidean or unitary space is *normal* if it commutes with its adjoint: $f \circ f^\dagger = f^\dagger \circ f$.

Remark 14.1 (Dictionary). For $f \in \operatorname{End}(V)$, complex $c \in \operatorname{Mat}(n \times n, \mathbb{C})$, real $a \in \operatorname{Mat}(n \times n, \mathbb{R})$:

- self-adjoint $f^\dagger = f$ — hermitian $\bar{c}^t = c$ — symmetric $a^t = a$;
- anti-self-adjoint $f^\dagger = -f$ — skew-hermitian $\bar{c}^t = -c$ — skew-symmetric $a^t = -a$;
- unitary/orthogonal $f^\dagger = f^{-1}$ — unitary $\bar{c}^t c = \mathbf{1}$ — orthogonal $a^t a = \mathbf{1}$.

All these types are normal.

Theorem 14.1 (Spectral theorem, unitary case)**.** *Let V be a finite-dimensional unitary space and f normal. Then V has an orthonormal basis of eigenvectors of f; i.e. f is described by a diagonal matrix in a suitable ONB. Conversely, any endomorphism diagonalizable by an ONB is normal.*

Sketch. *Induction on n: a normal f has an eigenvector v (a root of χ_f exists over $\mathbb{C}$); its orthogonal complement is invariant under both f and $f^\dagger$, and f stays normal there. Key lemma: a normal* triangular *matrix is already diagonal, as comparing the diagonal of c^*c and cc^* shows.*

Lemma 14.1. *For normal f: eigenvectors belonging to distinct eigenvalues are orthogonal.*

Theorem 14.2 (Spectral characterization)**.** *For normal f on a finite-dimensional unitary space:*

- *f self-adjoint $\iff$ all eigenvalues of f are real;*
- *f anti-self-adjoint $\iff$ all eigenvalues are purely imaginary;*
- *f unitary $\iff$ $|\lambda| = 1$ for every eigenvalue.*

Theorem 14.3 (Spectral theorem, real symmetric case)**.** *Let V be a finite-dimensional Euclidean space and f self-adjoint ($f^\dagger = f$; symmetric matrix model). Then V has an ONB of eigenvectors of f with* real *eigenvalues; a real symmetric matrix is diagonalizable by an orthogonal matrix: $u^{-1}au = u^t au$ diagonal.*

Theorem 14.4 (Normal form of orthogonal maps)**.** *For an orthogonal f on an n-dimensional Euclidean space there is an ONB in which f is described by a block diagonal matrix with blocks*

$$\begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \quad (t \neq 0), \text{ and diagonal entries } \pm 1.$$

Theorem 14.5 (Simultaneous diagonalization)**.** *Any family $\mathcal{F}$ of pairwise commuting normal endomorphisms of a finite-dimensional unitary space can be diagonalized simultaneously: there is one ONB of common eigenvectors.*

Corollary 14.1. *A real symmetric matrix with $a^r = 0$ for some r is zero.*

Theorem 14.6. *If the complex endomorphism f satisfies $f^r = \mathrm{id}$ for some r, then f is diagonalizable (over $\mathbb{C}$).*

Sketch. *Make V unitary with $\langle v, w\rangle' := \sum_{j<r}\langle f^j v, f^j w\rangle$; then f is unitary, hence normal, and $X^r - 1$ has simple roots.*

15 Quadratic Forms

Definition 15.1. A *quadratic form* on a real vector space is a map $q(x) = \langle x, x\rangle = x^t s x$ given by a symmetric bilinear form ($s \in \operatorname{Sym}(n, \mathbb{R})$ after a choice of basis). q is *positive definite* if $\langle\cdot,\cdot\rangle$ is.

Theorem 15.1 (Principal axes)**.** *For every quadratic form on an n-dimensional Euclidean space there is an ONB in which*

$$q(x) = \lambda_1 x_1^2 + \cdots + \lambda_n x_n^2$$

with real λ_j (diagonalize the symmetric matrix via the real spectral theorem).

Theorem 15.2 (Sylvester's law of inertia)**.** *There is a basis (not orthonormal, in general) in which*

$$q(x) = x_1^2 + \cdots + x_{r_+}^2 - x_{r_++1}^2 - \cdots - x_{r_++r_-}^2.$$

The pair (r_+, r_-), the signature*, does not depend on the basis; r_+ is the maximal dimension of a subspace on which q is positive definite, r_- the same for negative definite subspaces; $r_+ + r_- = \operatorname{rk} q$.*

Proposition 15.1 (Practical computation)**.** *Apply to s any elementary row operation together with the matching column operation; the result $u^t s u$ again represents q (in a new basis). Continuing like the Gaussian algorithm (scaling steps, not dividing by 0) yields the signature.*

Sketch. *Each double step is composition with an elementary matrix: row $\leftrightarrow$ left multiplication, matching column $\leftrightarrow$ right multiplication by its transpose.*

Example 15.1. For

$$s = \begin{pmatrix} -4 & 2 & 2 \\ 2 & -1 & 0 \\ 2 & 0 & -1 \end{pmatrix}$$

the first double step $u_{21}(\frac{1}{2})u_{31}(\frac{1}{2})$ (row, then the same on columns) clears the first row and column except the corner; continuing yields signature $(1, 2)$. Caution: all diagonal entries being negative does *not* imply negative definiteness.

Proposition 15.2 (Sylvester criterion)**.** *$s \in \mathrm{Sym}(n, \mathbb{R})$ is positive definite iff all* leading principal minors $\det(s_{ij})_{1 \le i,j \le k}$*, $k = 1, \dots, n$, are positive.*

Definition 15.2 (Minkowski space)**.** A symmetric bilinear form of signature $(1, n)$ on an $(n + 1)$-dimensional real space is a *Minkowski product*; in suitable coordinates $\langle x, x\rangle = x_0^2 - x_1^2 - \cdots - x_n^2$. Its isometry group is the Lorentz group $\mathrm{O}(1, n)$; vectors with positive "length squared" are timelike.

16 Quick Reference: Matrix Types & Groups

- *Symmetric* $a^t = a$: real eigenvalues, ONB of eigenvectors. *Skew-symmetric* $a^t = -a$: purely imaginary eigenvalues, 0 if n is odd.
- *Orthogonal* $u^t u = \mathbf{1}$: eigenvalues of modulus 1, $\det u = \pm 1$, $u^{-1} = u^t$; *rotation* if $\det = +1$.
- *Hermitian* $\bar{c}^t = c$, *unitary* $\bar{c}^t c = \mathbf{1}$, *normal* $\bar{c}^t c = c\bar{c}^t$: all diagonalizable by a unitary ONB (spectral theorem).
- *Triangular*: eigenvalues = diagonal entries. *Nilpotent* $a^r = 0$: only eigenvalue 0, strictly triangularizable. *Idempotent* $p^2 = p$: projection onto $\operatorname{img} p$ along $\ker p$; an orthogonal projection iff $p^2 = p = p^t$.
- Groups: $\mathrm{GL}(n, K) \supseteq \mathrm{SL}(n, K) = \ker \det$; $\mathrm{O}(n) \supseteq \mathrm{SO}(n)$; Sym_n with $\mathrm{sgn} : \mathrm{Sym}_n \to \{\pm 1\}$; $\det : \mathrm{GL}(n, K) \to K^\times$ is a homomorphism.
- Ranks: $\operatorname{rk} a = \dim$ of column span $= \operatorname{rk} a^t$; $\dim \ker f + \operatorname{rk} f = \dim V$; $\operatorname{rk}(g \circ f) \le \min\{\operatorname{rk} f, \operatorname{rk} g\}$.
- Dimensions: $\dim(S+T) + \dim(S \cap T) = \dim S + \dim T$; $\dim U + \dim U^\perp = \dim V$; $\dim U + \dim \operatorname{Ann} U = \dim V$.